

The background features a blue sky with white, stylized clouds. The clouds are composed of smooth, curved shapes in various shades of blue and white, creating a sense of depth and movement. The overall aesthetic is clean and modern.

Idaho National Laboratory

The U.S. lead laboratory for nuclear energy research and a premier center for national security, cybersecurity, advanced manufacturing, and energy systems

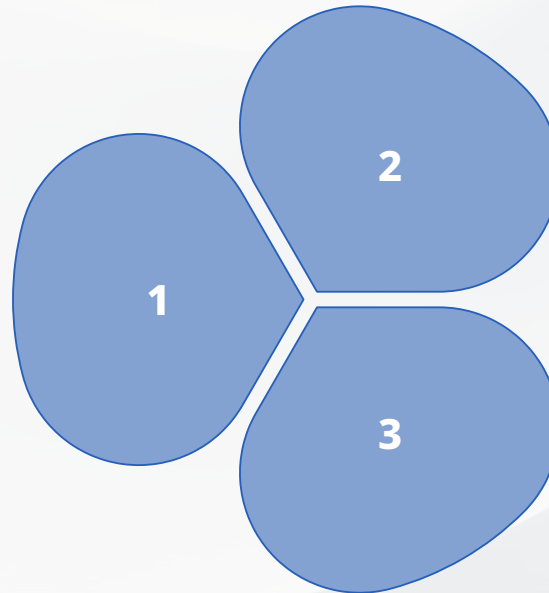
FACILITY 1

Control System Analysis

Control systems ensure that complex engineering systems behave safely, efficiently, and predictably. The core objective: **Measure** → **Compare** → **Correct**

Nuclear Reactor Control

Automated feedback ensures reactor power remains within safe operating bounds



Industrial Process Control

Continuous monitoring and correction of temperature, pressure, and flow

Autonomous Energy Systems

Self-regulating power systems integrating sensors, controllers, and actuators

Control Systems: Key Equations

Transfer Function

$$G(s) = \frac{Y(s)}{U(s)}$$

Input = $\mathbf{U}(s)$, Output = $\mathbf{Y}(s)$

PID Controller

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de}{dt}$$

P = proportional correction · I = accumulated error · D = prediction of future error. Used in nuclear reactors, chemical plants, aircraft, and robotics.

State Space

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

Models hundreds or thousands of interacting variables simultaneously.

Stability — Eigenvalues

$$\lambda_i < 0$$

All poles must remain in the **left half-plane** for a stable system. Without control systems: nuclear plants become unstable, aircraft cannot fly, electric grids collapse.

FACILITY 2

Separation Laboratory

One of chemical engineering's core disciplines — separating valuable materials from mixtures.



Uranium Purification

Isolating fissile material for nuclear fuel production



Rare Earth Extraction

Recovering critical materials for advanced technology applications



Water & Gas Separation

Purification for municipal systems, semiconductor manufacturing, and waste processing

Key Mathematics

Mass Balance: $Input = Output + Accumulation$

Component Balance: $Fz = Dx_D + Bx_B$

Fick's Law: $J = -D \frac{dC}{dx}$ — diffusion drives molecular transport

Separation Factor: $\alpha = \frac{K_A}{K_B}$ — higher α means easier separation

Materials & Fuel Complex / Radioactive Waste

Purpose

Study nuclear fuel behavior, structural alloys, corrosion, radiation damage, and radioactive waste immobilization to ensure safe storage, transportation, and reactor longevity.

Radioactive Decay

$$N = N_0 e^{-\lambda t}$$

Half-Life

$$t_{1/2} = \frac{\ln 2}{\lambda}$$

Heat Generation

$$Q = \lambda EN$$

Radiation Dose

$$D = \frac{E}{m}$$

Safe Storage

Engineered containment for radioactive waste

Transportation

Certified cask design and shielding standards

Reactor Longevity

Materials science to extend operational lifetimes

FACILITY 4

Electric Power Grid Test Bed

A 61-mile transmission loop that simulates the U.S. electrical grid under realistic operating conditions, studying cyberattacks, equipment failures, renewable energy integration, and grid resilience.

01

Real Power

$P = VI \cos \theta$ — the useful work delivered by the grid

03

Grid Frequency

$f = 60 \text{ Hz}$ — maintaining frequency is essential for grid stability

02

Reactive Power

$Q = VI \sin \theta$ — supports voltage stability across transmission lines

04

Power Flow & Ohm's Law

$P = \frac{V_1 V_2}{X} \sin \delta$ and $V = IR$ — govern transmission system stability and load distribution

Water Security Test

Protecting Water Systems Against Contamination, Cyber Threats & Biological Attacks



Continuity

$Q = Av$ — flow rate governs distribution system design



Bernoulli's Equation

$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$



Reynolds Number

$Re = \frac{\rho VD}{\mu}$ — determines laminar vs. turbulent flow

Applications

- Water treatment plant optimization
- Municipal distribution system monitoring
- Homeland security and contamination detection
- Chemical and biological attack response

Nuclear Fuels Test

Testing fuel performance under extreme temperatures, high radiation flux, and mechanical stress to support reactor safety and longevity.

Core Mathematics

Heat Conduction: $q = -k\nabla T$ — governs thermal management in fuel rods

Thermal Expansion: $\Delta L = \alpha L \Delta T$ — predicts dimensional change under heat

Fuel Burnup: $BU = \frac{\text{Energy}}{\text{Fuel}}$ — measured in GWd/tU; determines fuel lifecycle

Applications

- Extending reactor operational lifetime
- Improving safety margins
- Enabling Small Modular Reactors (SMRs)

Wireless End-to-End Testing

Reliable Communications for Military, First Responders, Satellites & Industrial Facilities

Shannon Capacity

$$C = B \log_2(1 + SNR)$$

Defines the theoretical maximum data rate achievable over a communications channel.

Friis Transmission Equation

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2$$

Calculates received signal power as a function of distance, antenna gain, and wavelength.

Signal-to-Noise Ratio

$$SNR = \frac{P_{signal}}{P_{noise}}$$

Applications

5G Networks

High-bandwidth, low-latency infrastructure resilience

Tactical Communications

Secure military and first responder links under adversarial conditions

Autonomous Vehicles

Reliable vehicle-to-infrastructure data exchange

Industrial IoT

Sensor networks for critical infrastructure monitoring

FACILITY 8

Advanced Test Reactor (ATR)

One of the world's most capable nuclear research reactors — testing fuels, reactor materials, medical isotope production, and naval reactor materials.

Neutron Flux

$$\phi = \frac{nv}{A}$$

Reactor Power

$$P = \Sigma_f \phi EV$$

Multiplication Factor

$$k = \frac{\text{Neutrons Produced}}{\text{Neutrons Lost}}$$

A critical reactor operates at exactly **k = 1**.

Applications

Space Reactors

Compact power systems for deep-space exploration

Naval Reactors

Propulsion materials testing for submarine and surface fleets

Medical Isotopes

Production of radioisotopes for cancer diagnostics and treatment

Advanced Fuels

Next-generation fuel forms for commercial and SMR reactors

FACILITY 9

Specialized Manufacturing Capabilities

Manufacturing nuclear components, advanced alloys, fuel assemblies, and robotic systems using additive manufacturing (3D printing).

Structural Mathematics

Stress: $\sigma = \frac{F}{A}$ — force per unit area in a material

Strain: $\epsilon = \frac{\Delta L}{L}$ — relative deformation under load

Young's Modulus: $E = \frac{\sigma}{\epsilon}$ — material stiffness

Fracture Mechanics: $K_I = Y\sigma\sqrt{\pi a}$ — predicts crack propagation before failure

Sectors Served

- Aerospace and Defense
- Nuclear energy components
- Semiconductor manufacturing

CROSS-CUTTING THEME

The Impact of AI Across INL Facilities

AI is transforming every one of these facilities by making experiments, monitoring, and decision-making faster and more predictive.



Control Systems

Reinforcement learning, anomaly detection, predictive control — improves safety and autonomous operation



Separation Labs

Process optimization, digital twins, molecular modeling — increases yield while reducing energy and waste



Nuclear Materials

Materials informatics, radiation damage prediction — accelerates discovery of radiation-resistant materials



Electric Grid

Grid forecasting, fault detection, cyberattack identification — builds more resilient, self-healing power systems



Water Security

Sensor fusion, contaminant detection, predictive maintenance — protects public health and critical infrastructure



ATR & Fuel Testing

Autonomous experiment planning and fuel performance prediction — shortens research cycles and reduces costly physical testing



Wireless Testing

AI-driven spectrum management and interference mitigation — enables reliable communications for defense and emergency response



Manufacturing

Machine vision, robotic inspection, generative design — higher-quality components with less waste and lower cost

Why This Matters to Society

INL Capabilities Support Many Sectors Beyond Nuclear Energy



National Defense

Secure communications, resilient infrastructure, and advanced materials for military applications.



Energy

Reliable nuclear power, smart electric grids, and integration of renewable energy sources.



Healthcare

Production of medical isotopes and safer sterilization technologies for hospitals and clinics.



Manufacturing

Stronger materials, additive manufacturing, and automated quality assurance for industry.



Environmental Protection

Improved radioactive waste management and water security for communities nationwide.



Artificial Intelligence

Digital twins, predictive maintenance, scientific machine learning, and autonomous control systems.

The broader trend is the convergence of engineering, physics, chemistry, cybersecurity, data science, and AI — augmenting engineers to accelerate research, improve safety, and enable data-driven decisions.